

Tezpur University
Sessional Test-1 Autumn 2026

Course Code: CSMT 502

Course Title: Selected Topics in Computer Networks

Max. Marks. 20 M

Time: 20 Min

Roll No: _____

Name: _____

I. Answer the following Multiple-Choice Questions.

(10 x 1 = 10 M)

1. In the OSI model, which combination correctly distinguishes the responsibilities of the Data Link and Network layers? []

- A. Data Link: process-to-process; Network: hop-to-hop
- B. Data Link: hop-to-hop; Network: source-to-destination
- C. Data Link: source-to-destination; Network: process-to-process
- D. Both provide only end-to-end delivery

2. Consider two packets belonging to the same message. They reach the destination through different routes and arrive out of order. Which switching mechanism most naturally permits this behavior? []

- A. Circuit switching
- B. Message switching
- C. Datagram packet switching
- D. Virtual-circuit packet switching

3. Consider two networks:

Network X: Average delay = 25 ms; successive packet delays vary from 24–26 ms.

Network Y: Average delay = 20 ms; successive packet delays vary from 5–50 ms.

For a delay-sensitive real-time application, which conclusion is most defensible based only on the performance concepts in the module? []

- A. Y must always perform better because its average delay is lower
- B. X may provide smoother delivery because its delay variation is substantially smaller
- C. Both must perform identically because throughput is unspecified
- D. X necessarily has greater bandwidth

4. Which statement correctly distinguishes throughput from bandwidth? []

- A. Throughput is always greater than bandwidth.
- B. Bandwidth represents achieved data rate, whereas throughput represents theoretical capacity.
- C. Bandwidth represents theoretical capacity, whereas throughput represents the actual achieved rate.
- D. They are necessarily identical for a congestion-free network.

5. In Go-Back-N, frames 0, 1, 2 and 3 are transmitted. Frame 1 is lost, while frames 2 and 3 reach the receiver. According to the behavior described in the module, what happens? []

- A. Frames 2 and 3 are buffered indefinitely.
- B. Only frame 1 is retransmitted.
- C. Frames 2 and 3 are discarded, and retransmission begins from frame 1.
- D. Frame 1 is ignored because later frames arrived.

6. Why can collisions still occur in CSMA even though a station senses the channel before transmission? []

[]

- A. Frames do not contain addresses.
- B. Propagation delay can cause two stations to perceive the channel as idle simultaneously.
- C. CSMA does not permit retransmission.
- D. Carrier sensing works only after a collision.

7. What is the principal advantage of piggybacking in full-duplex communication? []

- A. It eliminates error detection.
- B. It increases the frame sequence-number space.
- C. It combines acknowledgment information with outgoing data, reducing separate ACK frames.
- D. It eliminates the need for flow control.

8. Which situation can increase jitter without necessarily changing the physical propagation distance? []

- A. Replacing copper with a shorter cable
- B. Variable queuing delays experienced by successive packets
- C. Increasing signal propagation speed
- D. Reducing packet header size to zero

9. A TCP/IP host successfully determines the destination IP address from a domain name but still cannot deliver the application data to the correct process on the destination host. Which addressing information is most directly relevant to the unresolved problem? []

- A. MAC address
- B. IP address
- C. Port address
- D. DNS authoritative-server address

10. An HDLC receiver sends RNR. What should the sender infer? []

- A. The receiver has detected a CRC error.
- B. The receiver is currently not ready to receive further frames.
- C. The receiver requests selective retransmission.
- D. The receiver requests retransmission using Go-Back-N.

II. Answer the following fill in the blank questions. (5 x 1 = 5 M)

- 11. The variation in packet arrival delay over time is called _____.
- 12. A switching technique in which the complete message is stored at an intermediate node before being forwarded is called _____ switching.
- 13. The HDLC field used for CRC-based error detection is called the _____.
- 14. The maximum normalized throughput of Slotted ALOHA is approximately _____, compared with _____ for Pure ALOHA.
- 15. The retransmission-based error-control mechanism used in noisy-channel protocols is known as _____.

III. Answer the following True/False questions. (5 x 1 = 5 M)

- 16. The transport layer performs flow control only between two directly connected nodes.

17. In circuit switching, routing decisions are normally required for every unit of data after the dedicated circuit has been established.
18. HDLC asynchronous balanced mode gives both stations equal responsibility for controlling the link.
19. Carrier sensing and collision detection are different functions; therefore, a station can sense an idle medium and still subsequently experience a collision.
20. In Go-Back-N, successful physical arrival of a frame at the receiver does not necessarily imply that the frame will be accepted.